HAT-P-36b

R-0968

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-36_230127 · created 2026-07-29T16:59:38+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459971.9713 +/- 0.0019 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1151 +/- 0.0094
Transit depth (Rp/Rs)^2	1.33 +/- 0.22 [%]
Orbital Inclination (inc)	84.21 +/- 1.57
Airmass coefficient 1 (a1)	0.9709 +/- 0.0069
Airmass coefficient 2 (a2)	0.028 +/- 0.034
Scatter in the residuals of the lightcurve fit is	0.7 %
Best Comparison Star	#3 - [544, 136]
Optimal Aperture	4.04
Optimal Annulus	7.88
Transit Duration (day)	0.0885 +/- 0.0047

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1151 ± 0.0094 vs 0.126 (deviation 1.11σ)
Duration fit vs published	0.0885 d vs 0.0925 d (deviation 0.82σ)
Mid-time O–C	-3.0 min
Depth SNR	11.7
Residual scatter	0.7 %

Light curve

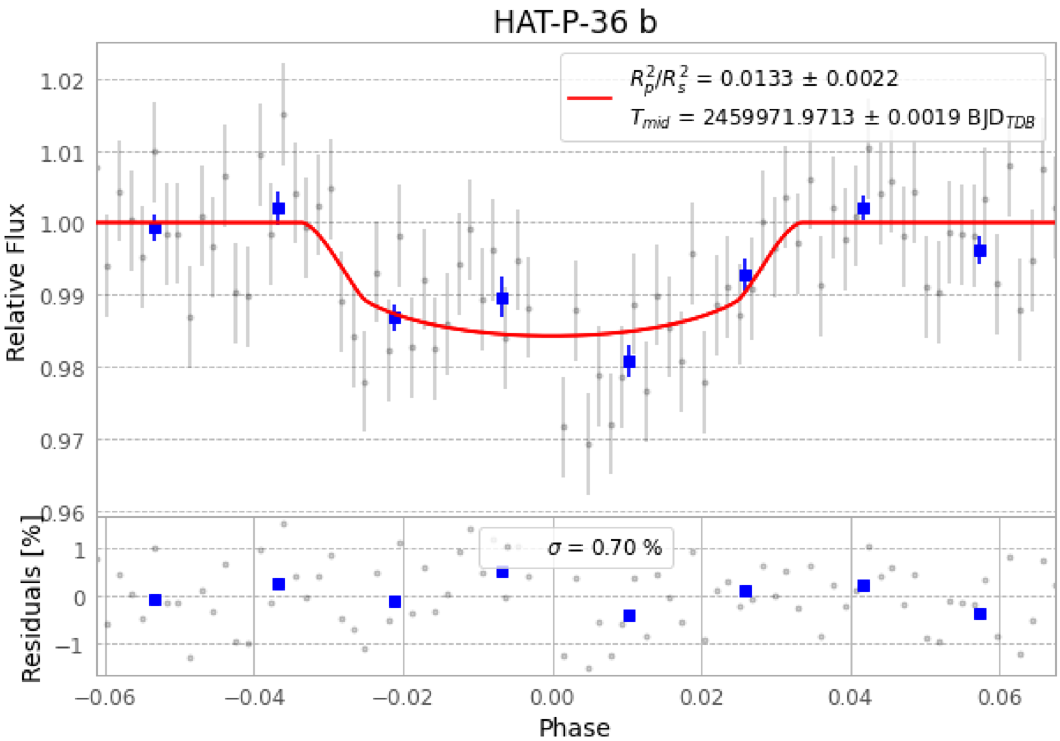

Figure 1 — FinalLightCurve_HAT-P-36 b_2023-01-27.png (SHA-256 870968f79d7104f7…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [551, 152], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 351d74e2eb68b828…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9`

Data provenance

83 FITS frames (55 MB), `HATP-36230127091517.FITS` → `HATP-36230127132116.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-36-230127.log (d7c8ed4988cd…), FinalLightCurve_HAT-P-36 b_2023-01-27.png (870968f79d71…), FinalLightCurve_HAT-P-36 b_2023-01-27.csv (b34656ed6ca8…)

Compute resources

Wall time 135.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)